National University of Modern Languages, Islamabad

Faculty of Engineering and Computing
Department of Computer Science

Third Assignment

BS CS/AI FALL-2025

Parallel and Distributed Computing

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Assignment Submission Date: 22-12-2025

Points: 10

Instruction:

- **Copied from any source leads to zero marks**
- **Hand written assignment is required.**

Learning Objectives

- Upon completion of this assignment, students will be able to:
- Analyze performance differences between CPU-based and GPU-based (CUDA) implementations
- Decompose CUDA program execution into host-side and device-side components
- Analyze memory access patterns and performance bottlenecks in CUDA programs
- Examine different failure types in distributed systems
- Compare fault-tolerance mechanisms and analyze their impact on system performance and reliability

Problem Statement

1. You are required to analyze a computational problem (e.g., vector addition or matrix multiplication) using both CPU and GPU (CUDA) implementations.
    - CPU vs CUDA Performance Analysis (6 Marks)
    - Implement the given computation using:
        - A sequential CPU program
        - A CUDA kernel-based GPU program
    - Analyze and compare the performance of both implementations.
2. Analyze the role of CUDA memory in your implementation.
    - Global vs shared memory usage
    - Memory access patterns
    - Impact of block and grid dimensions on performance
3. Analyze potential performance bottlenecks in your CUDA program.
    - Kernel launch overhead
    - Memory transfer latency
    - Thread divergence and occupancy
4. Consider a distributed system executing a parallel application.
    - Identification of possible failure types (node, process, network)
    - Impact of failures on correctness and performance
    - Classification of failures (fail-stop, omission, Byzantine)
5. Analyze and compare two fault-tolerance mechanisms, such as:
    - Checkpointing vs Replication

- Active vs Passive replication
    - Analysis Focus:
        - Resource overhead
        - Recovery time
        - Performance trade-offs

6. Analyze the trade-off between performance and reliability in fault-tolerant distributed systems.
    - Effect of fault tolerance on latency and throughput
    - Scenarios where minimal fault tolerance is acceptable